

Data Analytics

Excel Assignment 1 – Data Exploration

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Project Description

In this assignment, I worked on a product dataset in Microsoft Excel. The dataset has information about different products like laptops, t-shirts, sneakers, and more. Each product has a Product ID, Product Name, Brand Name, Price, Quantity, and Category.

My work is to explore this data using Excel formulas. I had to find the total price, count the products, find the average price, check the highest and lowest prices, label products based on price, and also pull out information from the Product ID column.

Task 1 & 2 – Basic Calculations

SUM, COUNT, AVERAGE, MIN, MAX

For the first two tasks, I used simple Excel functions on the Price column to find basic information about the data. Here are the formulas I used and the answers I got:

Calculation	Formula I Used	Answer
Total price of all products	=SUM(D2:D35)	10,100
How many products are there	=COUNTA(B2:B35)	34
Average price of products	=AVERAGE(D2:D35)	297.05
Lowest price	=MIN(D2:D35)	30
Highest price	=MAX(D2:D35)	1,000

I selected the price column from D2 to D35 for all these formulas. The SUM function adds all prices together. COUNT tells me how many rows have a price value. AVERAGE divides the total by the count. MIN finds the smallest price and MAX finds the biggest price.

Note: I also noticed that 2 products are repeated in the dataset (HP Laptop and Samsonite Laptop Bag appear twice). So there are actually 32 unique products.

Task 3 – IF Function (Price Range)

For this task, I added a new column called Price Range in Column G. I used the IF function to label each product. If the price is 500 or more, it gets the label High Price. If the price is less than 500, it gets Standard Price.

The formula I used is:

=IF(D2>=500,"High Price","Standard Price")

I applied this formula from G2 all the way down to G35. Here is how it works:

Condition	Formula I Used	Result
Price is 500 or more	=IF(D2>=500,"High Price","Standard Price")	High Price
Price is less than 500	Same formula for all rows	Standard Price

Out of 34 products, 8 products got the High Price label and the remaining 26 products got the Standard Price label. All 8 high-priced products are from the Electronics category.

Task 4 – SUMIF and COUNTIF

In this task, I used two conditional functions — SUMIF and COUNTIF. These functions work only on rows that match a condition, not on all rows.

SUMIF adds the prices of products only if they belong to the Electronics category. COUNTIF counts how many products have a price that is less than 100.

Function	Formula I Used	Answer
Total price of Electronics only (SUMIF)	=SUMIF(F2:F35,"Electronics",D2:D35)	8,050
How many products cost less than \$100 (COUNTIF)	=COUNTIF(D2:D35,"<100")	11 Products

The Electronics category has the highest total price of \$8,050 out of \$10,100 total. That means Electronics takes up most of the price in this dataset. Also, 11 products are priced below \$100, which means they are cheaper or budget products.

Task 5 – Text Functions (LEFT, RIGHT, MID)

In this task, I used text functions to pull out specific parts from the Product ID column. The Product ID looks like this: 28-JAN-US. The first two characters are the day, the middle three characters are the month, and the last two characters are the country code.

New Column	What It Takes Out	Formula I Used	Example
Day (Column H)	First 2 characters	=LEFT(A2, 2)	"28" from 28-JAN-US
Country Code (Column I)	Last 2 characters	=RIGHT(A2, 2)	"US" from 28-JAN-US
Month (Column J)	4th to 6th characters	=MID(A2, 4, 3)	"JAN" from 28-JAN-US

For example, the Product ID 28-JAN-US gives us Day = 28, Month = JAN, and Country Code = US. I applied all three formulas from row 2 to row 35.

The new columns are Column H (Day), Column I (Country Code), and Column J (Month) in the Excel file.

Excel Screenshot

Dataset Summary		
Task 1 - Basic Calculations (SUM, COUNT, AVERAGE)		
CALCULATION	EXCEL FORMULA	ANSWER
Total Price of all products	(=sum(D2:D35))	10100
Count of Products	(=counta(B2:B35))	34
Average Price	(=average(D2:D35))	297.06
Note: I also noticed that 2 products are repeated in the dataset (HP Laptop and Samsonite Laptop Bag appears twice). So there are actually 32 unique products.		
Task 2 – Minimum and Maximum Price (MIN, MAX)		

CALCULATION	EXCEL FORMULA	ANSWER
Min Price	(=min(D2:D35))	30
Max Price	(=max(D2:D35))	1000
Task 3– IF Function (Price Range)		
(=if(D2>=500,"High Price","Standard Price"))		
Task 4 – Conditional Functions (SUMIF, COUNTIF)		
CALCULATION	EXCEL FORMULA	ANSWER
SUMIF – Total Price of Electronics	(=sumif(F2:F35,"Electronics",D2:D35))	8050
COUNTIF – Products with Price < \$100	(=countif(D2:D35,"<100"))	11
Task 5 – Text Functions (LEFT, RIGHT, MID)		
Column H: Day =LEFT(A2,2) Column I: Country Code =RIGHT(A2,2) Column J: Month =MID(A2,4,3)		

Conclusion

In this assignment, I learned how to use basic Excel functions to explore a product dataset. I used SUM, COUNT, AVERAGE, MIN, and MAX to understand the price data. I used the IF function to label products as High Price or Standard Price. I used SUMIF to find the total price of Electronics and COUNTIF to count products below \$100. Finally, I used LEFT, RIGHT, and MID to pull out useful information from the Product ID column.

This assignment helped me understand how Excel can be used to analyze data in a simple and organized way. All the formulas are saved in the Excel file and the answers are correct as per the dataset.